Preliminary Latency Estimation for MixSense

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1 Introduction

Latency is a critical consideration in interactive musical systems because excessive delay can reduce responsiveness, disrupt timing relationships between modalities, and weaken the sense of real-time control. For MixSense, latency is especially relevant because the system combines user input, digital processing, communication, and haptic feedback within a bidirectional audio-haptic interaction loop.

This document develops a preliminary latency model to **support risk prediction** and **design-requirement setting**. The representative samples and control platforms discussed in the following sections are **not proposed design choices or implementation plans.** Instead, they are selected as reference examples to provide **realistic lower- and upper-bound timing estimates** for major system components.

The resulting latency budget is intended to **improve understanding** of where delay may occur, identify components that could threaten the synchronization requirement, and provide a **quantitative basis** for later design decisions and prototype validation.

The current system-level requirement is:

$$\boxed{T_{\text{total}} \leq 40 \text{ ms}} \tag{1}$$

where T_{total} represents the measured delay from a user control event or relevant audio-state change to the onset of the corresponding haptic feedback.

2 Background Research

2.1 Latency in Interactive Musical Systems

Low and consistent latency is an established requirement in digital musical instruments and interactive music systems. McPherson, Jack, and Moro evaluated the latency of common microcontroller-to-computer interaction architectures and emphasized that total latency should be considered across the complete interaction chain rather than at a single component level [1].

Subsequent perceptual work by Jack et al. showed that action-to-sound latency and latency variation can reduce the perceived quality of digital musical instruments [2]. This motivates considering both average latency and timing variability (jitter) when evaluating Mixsense.

More recent literature continues to treat action-to-sound latency as a significant constraint in interactive musical systems. For example, Turchet et al. discuss approximately 10 ms as a commonly referenced action-to-sound design target for digital musical instruments and identify latency and jitter as important challenges in multimodal musical interaction [3].

2.2 Audio–Haptic Synchronization

Mixsense differs from a conventional digital musical instrument because the user receives more than one feedback modality. The relevant engineering concern is therefore not only absolute system latency, but also the temporal alignment between the auditory and haptic feedback paths.

One useful quantity is the audio–haptic synchronization error:

$$\Delta T_{\mathrm{A-H}} = \left| T_{\mathrm{control}\to\mathrm{audio}} - T_{\mathrm{control}\to\mathrm{haptic}} \right| . \tag{2}$$

Psychophysical work has shown that the perception of audio–tactile synchrony depends strongly on the task and on which modality leads the other [4]. Therefore, Mixsense will not assume a single universal perceptual threshold. Instead, the project will use the 40 ms system requirement as an engineering bound and experimentally characterize the actual end-to-end timing of the prototype.

2.3 Latency as an Accumulated Quantity

The preliminary Mixsense control-to-haptic latency can be decomposed as

$$\boxed{T_{\mathrm{total}} = T_{\mathrm{input}} + T_{\mathrm{MCU}} + T_{\mathrm{comm}} + T_{\mathrm{DSP}} + T_{\mathrm{driver}} + T_{\mathrm{actuator}}} \tag{3}$$

where:

- T_{input}: physical input acquisition and filtering;
- T_{MCU}: embedded processing and control update;
- T_{comm}: bidirectional communication;
- T_{DSP}: DAW/plugin/audio processing and block scheduling;
- T_{driver}: haptic-driver command/startup time;
- T_{actuator}: actuator electromechanical response to a detectable vibration level.

Equation 3 is intentionally modular so that individual terms can be updated as hardware and software choices are finalized.

3 Component-Level Latency Estimation

3.1 Input Controls and Sensors

Medeiros and Wanderley's [5] review classifies musical-interface sensing by physical quantity, including force/pressure, position/displacement, touch/proximity, and inertial sensing. For Mixsense, these quantities could be divided into control-oriented classes: continuous force, absolute position, relative incremental, discrete event, and gesture/motion inputs. The reorganized classes map more directly to latency prediciton, user actions, and hardware choices.

To estimate worst-case input latency, Table 1 summarizes one representative device from each control class and the main sources of delay. For each input devices, the worst-case input latency is

Table 1: Representative input-control latency considerations.

Input Type	**Representative Example**	**Sensor Response**	**Polling**	**Interface**	**Filtering / Debounce**
Force / Pressure	Interlink FSR 402 [6]	Not explicitly specified; to be measured	≤ 1 ms at 1 kHz	ADC	Filtering may dominate depending on smoothing settings
Absolute Position	Bourns PTB Slide Potentiometer [7]	Negligible relative to sampling	≤ 1 ms at 1 kHz	ADC	Optional low-pass or moving-average filtering
Discrete Input	Omron B3F Tactile Switch [8]	Mechanical contact response	≤ 1 ms at 1 kHz	GPIO	Contact bounce ≤ 5 ms
Relative / Incremental	Bourns PEC11R Rotary Encoder [9]	Mechanical contact response	≤ 1 ms at 1 kHz	GPIO / interrupt	Contact bounce ≤ 2 ms
Gesture / Motion	Bosch BMI270 IMU [10]	0.625 ms at 1.6 kHz accelerometer Output Data Rate (ODR)	Sensor-driven	I^2C / SPI	0.156 ms before filtering / communication

estimated as the sum of sensor response, polling interval, interface delay, and filtering or debounce delay.

$$T_{\text{input,worst}} = T_{\text{sensor,worst}} + T_{\text{poll,worst}} + T_{\text{interface,worst}} + T_{\text{filter,worst}}$$

Among the representative samples in Table 1, the mechanical push buttons provides the largest documented delay. A standard 1kHz polling rate [11] is considered here and all sections below.

$$T_{\text{poll,worst}} = \frac{1}{f_{\text{poll}}},$$

$$T_{\text{poll,worst}} = 1 \text{ ms}.$$

Thus, with a maximum contact-bounce time of 5 ms, the preliminary worst case input-control latency is approximately 6 ms. However, mechanical buttons do not usually transition cleanly from 0 to 1 once. When pressed, the contacts can rapidly bounce between states for a few milliseconds which leads MCU to interpret one physical press as several presses. For conservative purposes, a 5 ms software debounce buffer has been added here so the system will accept new state only after remaining stable for 10 ms.

$$T_{\text{button,worst}} = T_{\text{poll}} + T_{\text{bounce,max}} + T_{\text{debounce,margin}} = 1 + 5 + 5 = \boxed{11 \text{ ms}}$$

3.2 MCU

MCU latency mainly results from input sampling, local computation, task scheduling, and buffering before communication:

$$T_{\text{MCU-stage}} = T_{\text{acquisition}} + T_{\text{local processing}} + T_{\text{communication}}.$$

To predict latency for Mixsense MCU, Table 2 below lists five commercial micro-controller with different forms of timing information as per the datasheets. Please note that information below is not end-to-end latency specifications, so their actual control latency must be experimentally measured.

Table 2: Typical Commercial Micro-controllers.

Platform	**Processing Estimate**	**Communication**	**Notes**
ESP32-S3[12]	Dual-core Xtensa LX7 up to 240 MHz, 2× ADC.	USB 2.0, UART, SPI, I^2C, Wi-Fi, BLE 5.	USB speed: 12Mbit/s; At maximum ADC $f_s = 100$ kS/s, the raw sampling interval is 10 μs. Integrated ADC, but ADC sensing precision need filtering / calibration.
Bela[13]	BeagleBone-based real-time embedded audio-sensing platform.	USB, UART, SPI, I^2C, onboard I/O for analog, digital, and audio.	<1 ms minimum round-trip audio latency; audio sampling rate = 44.1 kHz; default block size = 16 samples; adjustable block size = 2–128 samples; analog sample rate = 22.05 kHz. Relative expensive.
Teensy 4.1[14]	ARM Cortex-M7 at 600 MHz, 1 MB RAM.	USB, UART, SPI, I2C, Ethernet.	USB speed: 480 Mbit/s, I2C up to 1Mbit/s, 18 analog inputs, support 3.3V only.
Raspberry Pi[15]	Quad-core Cortex-A76 at 2.4 GHz.	USB, Ethernet, WI-FI, BLE, SPI, I2C, UART.	Huge computational capability $f_cpu = 2.4GHz$; However, jitter exits for OS scheduling with external ADC required.
Arduino[16]	Renesas RA4M1 Cortex-M4 at 48 MHz; 14-bit ADC.	USB, UART, SPI, I2C, CAN	f_{clock}= 48MHz, 6 analog inputs up to 14 bits, 5V I/O.

Based on the available datasheets, Bela demonstrates a <1 ms real-time operation at the system level as a upper limit while others only indicate sampling intervals on the order of tnes of microseconds.

If we pick Arduino as a lower limit with a 1kHz polling rate for one control loop:

$$f_{\text{poll}} = 1 \text{ kHz} = 1000 \text{ samples/s}$$

Then the time between two polling instants (ex., physically performing two ADC reads) is

$$T_{\text{poll}} = \frac{1}{f_{\text{poll}}} = \frac{1}{1000} = 0.001 \text{ s} = \boxed{1 \text{ ms}}.$$

Therefore, a preliminary range of **0.1-1ms** is reasonable for MCU-local processing latency. Communication latency should be considered separately depends on Mixsense control algorithm as per example below,

$$f_{\text{arduino}} = 48 \text{ MHz},$$

So, one CPU clock cycle is

$$T_{\text{cycle}} = \frac{1}{48 \times 10^6} \approx 20.8 \text{ ns},$$

then hypothetically, N CPU cycles cost

$$T_{\text{compute}} = NT_{\text{cycle}} = \frac{N}{48 \times 10^6}.$$

3.3 Communication and Audio Processing

Software and communication delays arise primarily from DAW block processing, inter-thread data transfer, and USB transport. Unlike actuator latency, these delays are mainly determined by software configuration and communication scheduling.

For block-based audio processing, the buffer duration is

$$T_{\text{buffer}} = \frac{N}{f_s},$$

where N is the audio-buffer size and f_s is the sampling frequency. At a sampling rate of 48 kHz, common buffer sizes correspond to

$$\frac{128}{48000} \approx 2.67 \text{ ms},$$

$$\frac{256}{48000} \approx 5.33 \text{ ms},$$

and

$$\frac{512}{48000} \approx 10.67 \text{ ms}.$$

These values represent the duration of one audio-processing block rather than a guaranteed end-to-end DAW latency; actual timing may depend on when an event arrives relative to the processing boundary and on additional host buffering.

The handoff between the real-time audio thread and middleware is expected to contribute comparatively little delay. VST3 provides non-blocking data-exchange mechanisms intended for transferring information between processing and non-real-time contexts without interrupting the audio thread [17]. For the preliminary latency budget, this contribution is treated as approximately

$$T_{\text{handoff}} < 1 \text{ ms},$$

while acknowledging that operating-system scheduling may introduce additional variation outside the scope of the current component-level estimate.

USB communication introduces an additional transport interval. The USB MIDI 2.0 specification provides a full-speed interrupt-endpoint configuration with a 1 ms polling interval [18]. Therefore, a preliminary one-way USB transport allowance of

$$T_{\text{USB}} \approx 1 \text{ ms}$$

is used for the reference latency budget. This value represents the transport scheduling interval rather than complete host-side application latency.

Table 3 summarizes the software and communication terms considered in this analysis.

Table 3: Representative software-processing and communication latency contributions.

Latency Source	**Reference Delay**	**Basis**
DAW audio buffer, 128 samples	2.67 ms	128/48,000
DAW audio buffer, 256 samples	5.33 ms	256/48,000
DAW audio buffer, 512 samples	10.67 ms	512/48,000
Audio-thread / middleware handoff	< 1 ms	Preliminary engineering allowance; VST3 supports non-blocking data exchange [17]
USB full-speed transport	~ 1 ms	Full-speed interrupt-endpoint polling interval [18]

For the preliminary system-level estimate, using the 128-sample DAW buffer gives

$$T_{\text{software,min}} \approx 2.67 + 1 + 1 = \boxed{4.67 \text{ ms}},$$

while the 512-sample buffer gives

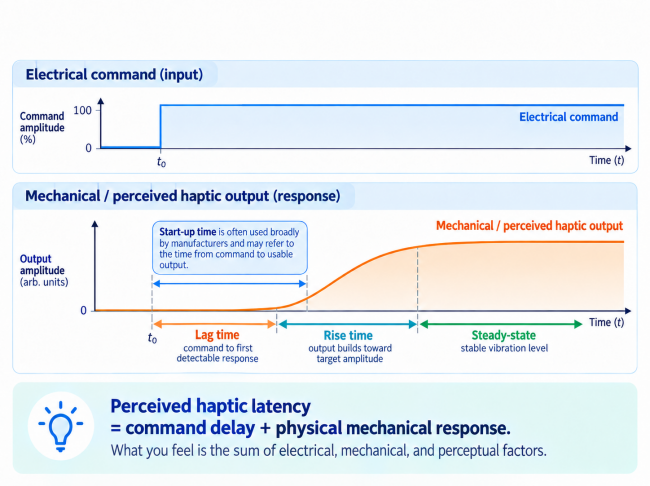
$$T_{\text{software,max}} \approx 10.67 + 1 + 1 = \boxed{12.67 \text{ ms}}.$$

These values are intended as reference bounds rather than measured implementation latency and should be validated experimentally once the final DAW, middleware, and USB architecture are established.

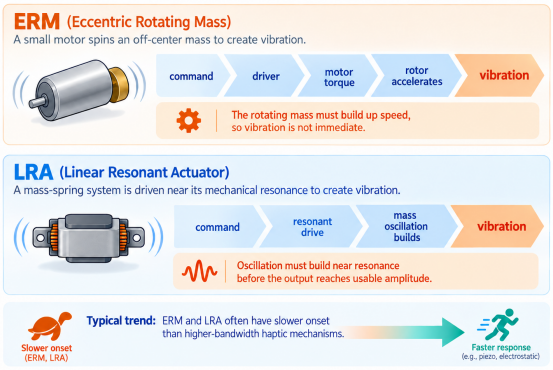
3.4 Haptic Feedback

Haptic feedback can be broadly delivered through tactile and kinesthetic mechanisms, including vibrotactile, force, resistance, and surface-based actuation.

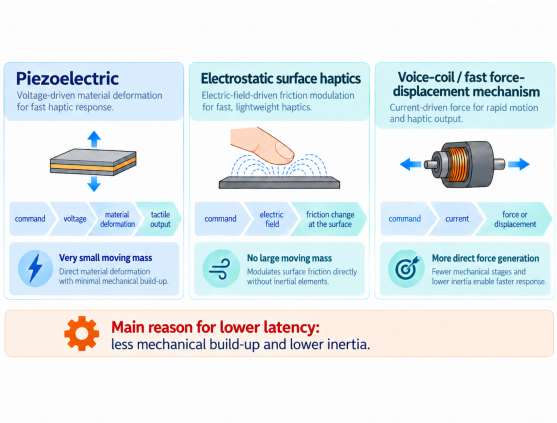
From a latency perspective, vibrotactile feedback requires particular attention because the actuator must undergo a physical mechanical response before the feedback becomes perceptible (Fig. 1a). Thus, a non-negligible delay can be introduced compared with some higher-bandwidth haptic mechanisms (Fig. 1b and c).



(a) Physical mechanical response in haptics.



(b) Mechanical build-up in ERM and LRA actuators.



(c) Higher-bandwidth haptic mechanisms.

Figure 1: Physical response characteristics of different haptic mechanisms: (a) electrical command and mechanical/perceived haptic response; (b) response build-up in ERM and LRA actuators; and (c) higher-bandwidth haptic mechanisms with reduced mechanical build-up.

In musical haptic interfaces, Armitage and Ng [19] identify actuator latency as an important factor affecting auditory-haptic synchronization. They characterized lag, ris, and stop times for representative ERM and LRM actuators. Motivated by this approach, section 3.4.1 analyzed the temporal responses of representative haptic actuators that relevant to Mixsense.

In addition to the actuator itself, a haptic driver is required to convert controller digital commands into the analog electrical excitation needed by the actuator. Its communication, switching, and waveform-generation delays form an additional part of the haptic output path and are therefore considered separately from the actuator response in Section 3.4.2.

3.4.1 Haptic Actuator

* TDK reports rapid start-up below 1 ms for the PowerHap piezoelectric actuator family.

Table 4: Representative haptic actuator response characteristics.

Actuator Type	**Representative Sample / Datasheet**	**Lag Time**	**Start Time**	**Rise Time**	**Fall / Stop Time**
ERM	Precision Microdrives 310-118.001, 10 mm coin vibration motor [20].	41 ms	–	96 ms	–
LRA	Precision Microdrives C12-003[21].	10 ms	–	≤32 ms	≤125 ms
Piezoelectric	TDK PowerHap 1313H018V120 [22].	–	< 1 ms*	–	–
Electrostatic Surface Haptics	Representative electrostatic / electrovibration surface-haptic system [23].	–	∼5 ms**	–	–
Voice-Coil / Fast Force-Displacement	Voice-coil actuator [23].	–	∼5 ms**	–	–

** Representative response-time values are taken from published haptic-actuator comparisons rather than a manufacturer-specified lag/rise measurement for the listed sample; therefore, they should be treated as technology-level reference values rather than device-specific guarantees.

Actuator latency is defined by the onset time, i.e, the delay from the drive command to the first usable haptic response. Thus, we consider start and lag time only for this analysis. As per Table 4, the worst documented onset is **41 ms** for a coin-ERM motor while LRA as another actuator to provide vibrotactile feedback has **10 ms** lag time.

$$\boxed{T_{\text{actuator,best}} \approx 1 \text{ ms}}$$

$$\boxed{T_{\text{actuator,worst}} \approx 41 \text{ ms}}$$

Beyond onset latency, rise time describes how quickly the haptic output builds toward its intended intensity after onset, while fall time describes how quickly the output decays after the drive command is removed. These vales are therefore treated as metics for dynamic response. Specifically, they affect pulse sharpness, continuous tracking, and how accurately changes in haptic intensity can follow the intended signal.

3.4.2 Haptic Driver

The selected driver examples cover themain actuator types considered for Mixsense, including ERM/LRA, piezoelectric, and force-feedback actuators, as well as simple MOSFET switch for direct drive. Since the driver contribution is expected to be much smaller than the actuator response, only the worst-case driver latency is carried forward in the system-level estimate. From Table 5,

Table 5: Representative haptic driver latency characteristics.

Driver Type	**Representative Sample / Datasheet**	**Compatible Actuator**	**Startup / Switching Time**	**Interface**
Dedicated ERM/LRA Driver	TI DRV2605L[24]	ERM / LRA	0.7 ms	I^2C, PWM, analog, GPIO trigger
Piezoelectric Haptic Driver	TI DRV2667[25]	Piezoelectric	2 ms	I^2C, PWM, analog
Direct MOSFET Switching	AO3400-class N-channel MOSFET[26]	ERM / simple DC actuator	μs-class	GPIO / PWM
H-Bridge / Current Driver	TI DRV8833[27]	Voice-coil / DC motor / force-feedback actuator	–	PWM / GPIO

the largest documented startup delay is around

$$\boxed{T_{\text{driver,worst}} \approx 2 \text{ ms}}$$

for the piezo driver.

4 MixSense System-Level Latency Estimation

4.1 Reference Architecture

The preliminary reference interaction path is

User Control $\rightarrow$ **MCU** $\rightarrow$ **USB / Host** $\rightarrow$ **DAW / Audio Effect** $\rightarrow$ **Feedback Message** $\rightarrow$ **MCU** $\rightarrow$ **Haptic Driver** $\rightarrow$ **Actuator**.

The following scenarios do not represent selected implementation configurations. Instead, they combine representative lower and upper latency bounds identified in Section 3 to estimate the range in which the MixSense end-to-end latency could plausibly fall.

For the bidirectional MixSense interaction, the audio and haptic feedback paths can be considered separately. Both paths initially share the input, MCU, forward communication, and DAW processing stages:

$$T_{\text{input}} + T_{\text{MCU}} + T_{\text{USB,fwd}} + T_{\text{buffer}} + T_{\text{handoff}}.$$

The haptic path additionally requires the processed feedback state to return to the embedded controller and drive the haptic hardware:

$$T_{\text{control}\to\text{haptic}} = T_{\text{control}\to\text{audio}} + T_{\text{USB,ret}} + T_{\text{driver}} + T_{\text{actuator}}.$$

The audio–haptic timing mismatch can therefore be expressed as

$$\Delta T_{\text{audio}-\text{haptic}} = \left| T_{\text{control}\to\text{haptic}} - T_{\text{control}\to\text{audio}} \right|,$$

which simplifies to

$$\boxed{\Delta T_{\text{audio}-\text{haptic}} = T_{\text{USB,ret}} + T_{\text{driver}} + T_{\text{actuator}}}$$

under the present reference architecture.

4.2 Best-Case Reference Scenario

The best-case scenario combines the lower-latency representative values identified in the component-level analysis. A simple continuous input is assumed with approximately 1 ms acquisition latency, MCU-local processing is represented by the lower bound of 0.1 ms, and a 128-sample DAW buffer is used at 48 kHz.

Because the interaction path is bidirectional, two 1 ms USB transport intervals are included. A DRV2605L-class dedicated haptic driver is used as a documented low-latency driver reference, and piezoelectric actuation provides the lower actuator-onset reference.

Table 6: Best-case reference latency scenario for MixSense.

Stage	**Reference Delay**	**Basis**
Input control	1.00 ms	Fast continuous control with 1 kHz polling
MCU-local processing	0.10 ms	Lower engineering estimate from Section 3.2
DAW audio buffer	2.67 ms	128 samples at 48 kHz
Audio-thread / middleware handoff	1.00 ms	Conservative upper allowance of < 1 ms
USB forward transport	1.00 ms	Full-speed interrupt polling interval
USB return transport	1.00 ms	Full-speed interrupt polling interval
Haptic driver	0.70 ms	DRV2605L startup reference
Haptic actuator onset	1.00 ms	Piezoelectric actuator reference
Estimated Total	**8.47 ms**	

Thus,

$$T_{\text{system,best}} = T_{\text{input,best}} + T_{\text{MCU,best}} + T_{\text{SW+comm,best}} + T_{\text{driver,best}} + T_{\text{actuator,best}} \tag{4}$$

$$= 1.00 + 0.10 + (2.67 + 1.00 + 1.00 + 1.00) + 0.70 + 1.00 \tag{5}$$

$$= \boxed{8.47 \text{ ms}}. \tag{6}$$

The estimated best-case latency is therefore well below the current 40 ms synchronization requirement.

4.3 Worst-Case Reference Scenario

The worst-case scenario combines the largest representative delays carried forward from the component-level analysis. The input stage uses the 11 ms mechanical-button estimate, including polling, contact bounce, and an additional software debounce margin. MCU-local processing is assigned the 1 ms upper engineering bound.

For software processing, the largest considered DAW buffer of 512 samples at 48 kHz is used. The same 1 ms handoff allowance and two 1 ms USB transport intervals are included. The largest documented driver startup time considered is 2 ms, while the representative coin ERM provides the largest documented actuator onset of 41 ms.

Table 7: Worst-case reference latency scenario for MixSense.

Stage	**Reference Delay**	**Basis**
Input control	11.00 ms	1 kHz polling + 5 ms contact bounce + 5 ms debounce margin
MCU-local processing	1.00 ms	Upper engineering estimate from Section 3.2
DAW audio buffer	10.67 ms	512 samples at 48 kHz
Audio-thread / middleware handoff	1.00 ms	Conservative upper allowance of < 1 ms
USB forward transport	1.00 ms	Full-speed interrupt polling interval
USB return transport	1.00 ms	Full-speed interrupt polling interval
Haptic driver	2.00 ms	DRV2667 startup reference
Haptic actuator onset	41.00 ms	Representative coin-ERM lag time
Estimated Total	**68.67 ms**	

Thus,

$$T_{\text{system,worst}} = T_{\text{input,worst}} + T_{\text{MCU,worst}} + T_{\text{SW+comm,worst}} + T_{\text{driver,worst}} + T_{\text{actuator,worst}} \tag{7}$$

$$= 11.00 + 1.00 + (10.67 + 1.00 + 1.00 + 1.00) + 2.00 + 41.00 \tag{8}$$

$$= \boxed{68.67 \text{ ms}}. \tag{9}$$

This reference scenario exceeds the current 40 ms synchronization requirement by approximately

$$68.67 - 40 = \boxed{28.67 \text{ ms}}. \tag{10}$$

The result indicates that the proposed latency requirement is achievable under lower-latency component and software configurations, but cannot be assumed across all plausible configurations. In particular, the actuator response and DAW buffer size account for a substantial portion of the worst-case latency budget.

4.4 Estimated MixSense Latency Range

Combining the two reference scenarios gives the preliminary system-level latency range

$$\boxed{8.47 \text{ ms} \lesssim T_{\text{MixSense}} \lesssim 68.67 \text{ ms}} \tag{11}$$

or approximately

$$\boxed{T_{\text{MixSense}} \approx 8.5\text{–}68.7 \text{ ms}}. \tag{12}$$

This range should not be interpreted as a guaranteed implementation specification. It represents a preliminary risk envelope constructed from representative component and configuration bounds. The analysis suggests that the largest opportunities for reducing end-to-end latency are the selection of a faster haptic actuator and the use of a smaller DAW audio buffer. The final system latency should therefore be verified experimentally after the hardware, communication architecture, DAW configuration, and haptic-feedback implementation are finalized.

5 Open Design Decisions

The following items should be updated as Mixsense moves from proposal to implementation:

- Final definition of the 40 ms requirement: control-to-haptic, audio-state-to-haptic, or both.
- Maximum acceptable audio–haptic mismatch $\Delta T_{\text{A}-\text{H}}$.
- Required control sampling frequency.
- Audio buffer size and sample rate.
- USB protocol and message architecture.
- Exact haptic driver.

- ERM versus LRA actuator selection.
- Threshold used to define physical "haptic onset."
- Maximum acceptable jitter.

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